DGMS 2019 • SECTION K: ROCK MECHANICS & GEOTECHNICAL TESTING

Question 84: The Brazilian Test: Indirect Tensile Strength Determination

Official Examination Guide • Diametral Compression, Elastic Theory & Tensile Splitting Mechanics

OFFICIAL EXAMINATION QUESTION

The Brazilian test is used to determine

(a) Crushing strength	**(b) Tensile strength** CORRECT ANSWER
(c) Transverse strength	(d) Shear strength

1. Executive Summary & Core Principle

In experimental rock mechanics (ISRM Suggested Methods, Bieniawski & Hawkes 1978; ASTM D3967), determining the **tensile strength (σ_t)** of rock by direct pull is experimentally difficult because clamping rock core ends inevitably induces parasitic bending stresses and stress concentrations, causing specimen failure at the grips rather than in tension within the gauge section.

The **Brazilian Test (Indirect Tensile Strength Test)** overcomes this challenge by subjecting a circular cylindrical rock disc (diameter D, thickness t) to diametral line compressive loading (P):

- According to the classical theory of elasticity for a disc under concentrated diametral loads (Hondros 1959), a nearly uniform, horizontal **tensile stress (σ_t)** is generated across approximately 75% of the vertical loaded diameter.

- Because intact rock is far weaker in tension than in compression (compressive-to-tensile strength ratio $\sigma_c / \sigma_t \approx 10\text{ to }20$), the disc splits cleanly in pure tension along the vertical diametral line.

The indirect tensile strength (σ_t or σ_B) is calculated directly from the peak compressive failure load (P) via the formula:

```
σt = 2P / (π · D · t)
```

Therefore, the Brazilian test is specifically designed and used to determine **(b) Tensile strength**.

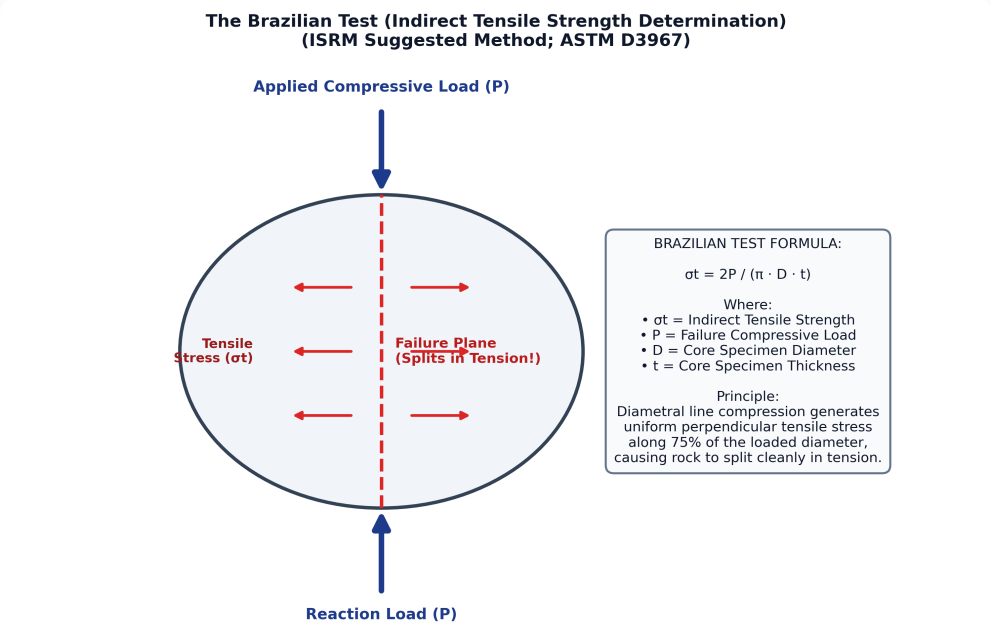


Figure 84.1: Specimen geometry, loading configuration, and internal tensile stress distribution during a standard Brazilian Test (ISRM Suggested Method; ASTM D3967).

2. Mathematical Formulation & Elastic Stress Distribution

For an elastic, homogeneous, isotropic circular disc of diameter D and thickness t compressed across a diameter by line load P, the stresses along the loaded diameter (y-axis) are given by:

```
Horizontal Stress (Tensile):
σx = (2P / π D t) · [ (1 − 4y²/D²) / (1 + 4y²/D²)² ]

At the Center of the Disc (y = 0):
σx = σt = 2P / (π · D · t) [Uniform Horizontal Tension]

Vertical Stress (Compressive):
σy = − (2P / π D t) · [ (3 + 4y²/D²) / (1 − 4y²/D²)² ]
```

Throughout the central region ($-0.75 R < y < +0.75 R$), σ_x is positive (tensile) and virtually constant. When this tensile stress reaches the intrinsic tensile strength of the rock material, Griffith tensile microcracks propagate vertically from the center outward, splitting the disc into two halves.

3. Comparison of Rock Strength Testing Methods

Strength Property	**Standard Laboratory Test**	**Loading Mode & Specimen Geometry**	**Key Governing Parameter**
Tensile Strength (σ_t)	**Brazilian Test (ISRM / ASTM D3967)**	Diametral line compression on thin disc ($t/D \approx 0.5$)	**$\sigma_t = \frac{2P}{\pi D t}$ (Option b — Target)**
Crushing / Compressive Strength (σ_c)	Uniaxial Compressive Strength (UCS)	Axial compression on cylindrical core ($L/D = 2.0\text{--}2.5$)	$\sigma_c = P / A$; measures bulk crushing resistance.
Transverse / Flexural Strength (σ_f)	Four-point / Three-point Bending Test	Transverse bending load applied on rock prism beam	Modulus of rupture $R = \frac{3PL}{2bd^2}$.
Shear Strength (τ)	Direct Shear Box / Triaxial Test	Shear displacement across predefined	Cohesion (c) and Internal

Strength Property	Standard Laboratory Test	Loading Mode & Specimen Geometry	Key Governing Parameter
		plane or Mohr-Coulomb failure envelope	Friction Angle (ϕ).

4. Systematic Option Evaluation

- **(a) Crushing strength: INCORRECT.** Crushing (compressive) strength is measured using the Uniaxial Compressive Strength (UCS) test on core cylinders ($L/D = 2.0\text{--}2.5$).
- **(b) Tensile strength: CORRECT ANSWER.** The Brazilian test is the international standard method for measuring the indirect tensile strength of rock.
- **(c) Transverse strength: INCORRECT.** Transverse strength (modulus of rupture) is determined by beam bending tests.
- **(d) Shear strength: INCORRECT.** Measured by direct shear box tests or triaxial compression cell testing.

> **Key Takeaway:** Brazilian Test = Indirect Tensile Strength ($\sigma_t = \frac{2P}{\pi D t}$). Hence, **(b) Tensile strength** is the definitive answer.

← Previous: Q.83 Arch Dam Load Transmission **Master Portal Index** **Next: Q.85 Tunnel Site Selection & Geological Strata →**